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Predict demand. Prevent waste.

AI-based forecasting for smart energy management

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Every forecasting error has a price.

In an energy market shaped by volatility, regulation and the renewable transition, the cost of getting demand wrong has never been higher.

Money on the table

Every over- or under-forecast adds dispatching cost. Multiplied across thousands of households, the errors compound.

Blackout risk

Underestimated evening peaks force expensive real-time balancing. Overload risk climbs.

Renewable lock-out

Solar and wind can only be matched to demand if that demand is known in advance.

OUR LABORATORY

2 M+
minute-level measurements

47
months of real data

34 K
hourly points after resampling

4
years of seasonality

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Three contenders. One arena.

Same dataset. Same protocol. Same evaluation window. The only thing that changes is the model — so the comparison is decided by the algorithm alone, not by the setup.

THE BASELINE

Seasonal Naïve

yesterday at the same hour

The floor. If a model can't beat copying yesterday, it isn't forecasting — it's noise.

THE VETERAN

SARIMA

(2,0,1) × (1,1,1,24)

Five parameters. Built for exactly this kind of seasonal signal.

THE FAVOURITE

LSTM

lookback 48 · hidden 64

Modern recurrent network. The market's default. Should win by reputation alone.

The veteran won.

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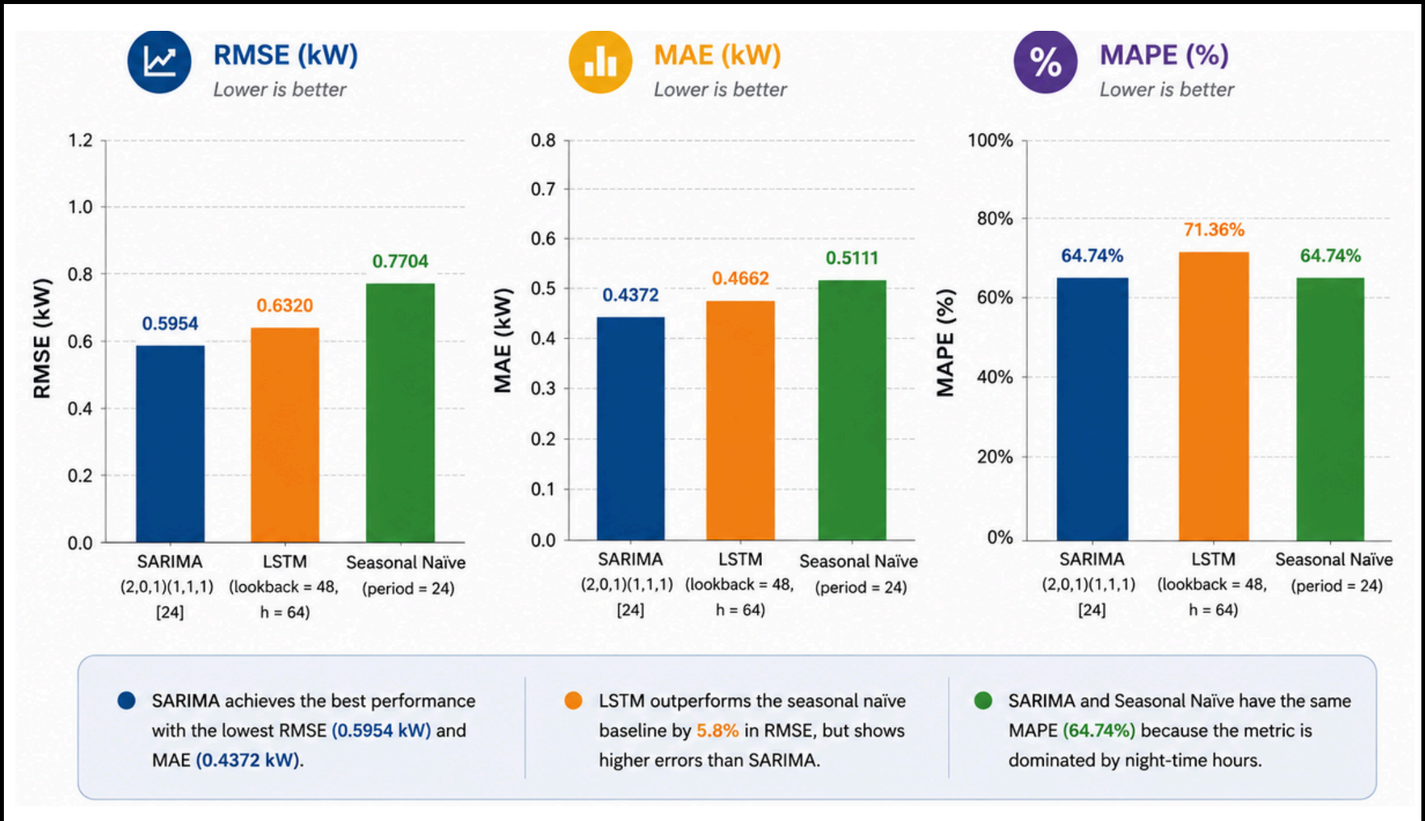

WINNER

SARIMA

Best across RMSE and MAE

0.5954

kW · RMSE on the held-out test set

-22.7 % beating the naïve baseline
-5.8 % beating a 64-unit LSTM

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From research to grid.

What this delivers, and what comes next.

Better accuracy. Cheaper to train. Easier to explain.

Five interpretable parameters. Hourly RMSE of 0.5954 kW — about 40 % of mean consumption — accurate enough for day-ahead scheduling and peak balancing.

Strategically aligned with Enel's roadmap.

Auditable, interpretable forecasts plug directly into renewable-demand matching and active demand management — the core of the decarbonisation agenda.

A clear path to keep widening the lead.

Next: exogenous regressors (temperature, calendar), probabilistic forecasts (prediction intervals) and extension to multi-household hierarchical settings.

Thank you.

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